

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 5: Arithmetic Progressions

Previous year board questions, topic-wise: 10 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Common difference and the nth term

- Find the common difference of the AP $\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, 4\sqrt{2}, \dots$ (a) $\sqrt{2}$ (b) 1 (c) $2\sqrt{2}$ (d) $-\sqrt{2}$
CBSE 2026, 1 mark
- The numbers $p - 1, p + 1$ and $2p + 3$ are in AP. Find p . (a) -2 (b) 4 (c) 0 (d) 2 CBSE 2023, 1 mark

2 Case study

- In a potato race, 10 potatoes lie in a straight line. The first is 5 m from a bucket at the starting point and the rest are 3 m apart. A runner starts at the bucket, fetches the nearest potato, runs back to drop it in the bucket, and repeats this for each potato in turn. (i) How far does the runner go to collect the first potato? (ii) How far to collect the second? (iii) Find the total distance run, OR the time taken for the whole race at an average speed of 5 m/s.
CBSE 2026, 4 marks
- In a fund-raising run, the first round is 300 m long and every round after it is 50 m longer than the one before, so the rounds are 300 m, 350 m, 400 m and so on. There are 10 rounds in all. (i) Write the 4th, 5th and 6th terms of this AP. (ii) How long is the 8th round? (iii) Find the total distance covered in all 10 rounds, OR the total distance covered by a runner who stops after 6 rounds. CBSE 2025, 4 marks

3 Sum of the first n terms

- The sum of the first m terms of an AP is $2m^2 + 3m$. Find its second term. CBSE 2025, 1 mark
- An A.P. has first term $a = 7$ and n th term $a_n = 84$, and its first n terms add up to $\frac{2093}{2}$. Find n .
(a) 22 (b) 24 (c) 23 (d) 26 CBSE 2024, 1 mark
- In an A.P., the first and eighth terms have sum 32 and product 60. Find the first term and the common difference, and then the sum of the first 20 terms. CBSE 2024, 5 marks
- An A.P. has 40 terms. Its first 9 terms add up to 153 and its last 6 terms add up to 687. Find the first term, the common difference and the sum of all 40 terms. CBSE 2024, 5 marks
- In an AP, the 11th term and the 17th term are in the ratio 3 : 4. Find the ratio of the 5th term to the 21st term, and the ratio of the sum of the first 5 terms to the sum of the first 21 terms.
CBSE 2023, 5 marks

10. 250 logs are stacked in rows: 22 in the bottom row, 21 in the next, 20 in the one above, and so on. How many rows are there, and how many logs are in the top row? **CBSE 2023, 5 marks**

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